Chapter 1

Concrete Groups

A **binary operation on a set G** is a rule $*$ which associates with every pair of elements a and b of **G** a unique element $a*b$ of **G**. More formally, a binary operation $*$ on **G** is a function $*: \mathbf{G}\times\mathbf{G}\to\mathbf{G}$ from the cartesian product

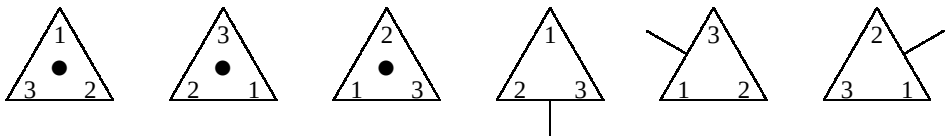
$$\mathbf{G}\times\mathbf{G} := \{(a,b) \mid a,b\in\mathbf{G}\}$$

into **G**. Note that ":=" means that the expression on the left is being defined as an abbreviation for the expression on the right. We write "$a*b$" to denote the image $*(a,b)$ of (a,b) under $*$. Abstract algebra has primarily grown out of the study of binary operations, and of the features common to many familiar binary operations. Binary operations abound in mathematics, as the examples below illustrate.

1. **Addition** $+$, **subtraction** $-$ and **multiplication** $\cdot$ are binary operations on the set **R** of real numbers and, suitably restricted, on the set **Z** of integers. Notice that **division** $\div$ is not a binary operation on either **R** or **Z** since $a\div b$ is not defined when $b=0$.

2. Let S be a set and let $P(S)$ denote the collection of subsets of S, called the **power set** of S. Then **intersection** $\cap$, **union** $\cup$, and **set difference** $\sim$ are binary operations on $P(S)$. Another important binary operation on $P(S)$ is the **symmetric difference**, $\oplus$, defined as
$$A\oplus B := \{x\in S \mid x \text{ is in either } A \text{ or } B \text{ but not both.}\},$$
that is, $A\oplus B = (A\cup B)\sim(A\cap B)$.

3. Let $\mathbf{S}_3$ denote the set of **permutations** of the set $\{1,2,3\}$, that is, one-to-one and onto functions from $\{1,2,3\}$ to itself. **Composition** $\circ$ forms a binary operation on $\mathbf{S}_3$ since the composition of two permutations is again a permutation. The set $\{1,2,3\}$ has exactly 6 permutations, illustrated below as rigid motions of an equilateral triangle.



$e=\left(\begin{smallmatrix}1&2&3\\1&2&3\end{smallmatrix}\right)$ $a=\left(\begin{smallmatrix}1&2&3\\2&3&1\end{smallmatrix}\right)$ $b=\left(\begin{smallmatrix}1&2&3\\3&1&2\end{smallmatrix}\right)$ $c=\left(\begin{smallmatrix}1&2&3\\1&3&2\end{smallmatrix}\right)$ $d=\left(\begin{smallmatrix}1&2&3\\3&2&1\end{smallmatrix}\right)$ $f=\left(\begin{smallmatrix}1&2&3\\2&1&3\end{smallmatrix}\right)$

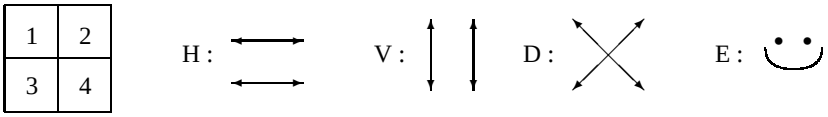
Here a permutation g is represented by putting corner x in the $g(x)$ position, for $x=1,2,3$. For example, a is achieved by starting in the e configuration and rotating

the triangle clockwise 120 degrees. The product $a \circ f = d$ is realized by doing f first and then a: we flip the triangle over the diagonal through corner 3 (f), and then rotate it clockwise 120 degrees (a), as we have

$$(a \circ f)(1) = a(f(1)) = a(2) = 3 = d(1)$$
$$(a \circ f)(2) = a(f(2)) = a(1) = 2 = d(2)$$
$$(a \circ f)(3) = a(f(3)) = a(3) = 1 = d(3).$$

4. The set $\mathbf{Z}_{12} = \{0,1,2,3,4,5,6,7,8,9,10,11\}$ denotes the numbers on a clock, where 0 replaces 12. We define $a +_{12} b$ to be the hour that is b hours after hour a. For example, $10 +_{12} 7 = 5$ since 7 hours after 10 o'clock is 5 o'clock.

5. Consider a 2×2 "chess board" on which you are allowed to make four different moves:



H move horizontally (1 to 2, 2 to 1, 3 to 4 or 4 to 3),

V move vertically (1 to 3, 3 to 1, 2 to 4 or 4 to 2),

D move diagonally (1 to 4, 4 to 1, 2 to 3 or 3 to 2), or

E stay in place.

Let $\mathbf{K}_4 = \{E,H,V,D\}$ and let $X * Y$ be the move achieved by doing Y first and then doing X.

6. Recall that the **determinant** of a 2×2 matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is the number $|A| := ad - bc$. Let $\mathbf{M}_1$ be the set of 2×2 matrices with determinant 1 under the usual matrix multiplication $\bullet$ defined by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \bullet \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ap+br & aq+bs \\ cp+ds & cq+ds \end{pmatrix}$$

A **binary system** consists of a set $\mathbf{G}$ together with a binary operation $* : \mathbf{G} \times \mathbf{G} \to \mathbf{G}$ on $\mathbf{G}$. If $\mathbf{G}$ is a finite set with n elements, we can fully define a binary operation $*$ with an n-by-n table. We list the elements of $\mathbf{G}$ as labels down the left side of the table and again across the top. Then the entry in the row labeled x and column labeled y is the value of $x * y$.

Constructing tables for a few sample finite binary systems provides a good opportunity to build some concrete intuition about their behavior before we begin to study them abstractly. We will make frequent reference to the examples below in the future.

Problem 1. *Using a separate sheet for each one, construct a table for the following binary systems:*

(i) $\mathbf{K}_4 = \{E,H,V,D\}$ *with* $*$,

(ii) $\mathbf{Z}_{12}$ *with* $+_{12}$*, the clock arithmetic,*

(iii) $\mathbf{P}_{abc}$ *with* $\oplus$*, the system of 8 subsets of the set* $\{a,b,c\}$ *under symmetric difference, and*

(iv) $\mathbf{S}_3$ *with* $\circ$*, the rigid motions of an equilateral triangle. (It will help to use a real triangle with a* $\bullet$ *on the top side!*)

Many operations have special properties that enhance our ability to do computations within them. A binary operation $*$ is **associative** if

$$a*(b*c) = (a*b)*c \quad \text{for every } a,b,c \in \mathbf{G}.$$

If $a,b,c \in \mathbf{G}$, then the expression "$a*b*c$" is in general ambiguous since it could mean either $a*(b*c)$ or $(a*b)*c$. When we know that $\mathbf{G}$ is associative, we generally omit the parentheses since $a*b*c$ will have the same value whichever way we interpret it.

We say that $\mathbf{G}$ is **commutative** if

$$a*b = b*a \quad \text{for every } a,b \in \mathbf{G}.$$

An element $e \in \mathbf{G}$ is an **identity** for $*$ if

$$e*a = a*e = a \quad \text{for every } a \in \mathbf{G}.$$

If e is an identity for $*$ and $a \in \mathbf{G}$, then an **inverse** for a is an element $b \in \mathbf{G}$ such that

$$a*b = b*a = e.$$

Our next definition isolates the most essential properties of binary systems.

Definition 1. *A* **group** *is a binary system* $\mathbf{G}$ *with* $*$ *for which*

(i) $*$ *is associative,*

(ii) *there is an identity element* $e \in \mathbf{G}$ *for* $*$*, and*

(iii) *each element of* $\mathbf{G}$ *has an inverse.*

For example, we can check that each of the binary systems of Problem 1 is a group. We call $\mathbf{K}_4$ the **Klein 4-Group**, $\mathbf{Z}_{12}$ the **group of integers modulo 12** and $\mathbf{S}_3$ the **symmetric group on 3 elements**. The fact that $\mathbf{P}_{abc}$ is associative is not obvious from the definition, but is a straightforward exercise in elementary set theory.

It turns out that most of what we will want to do does not require the commutative property. If the binary operation of a group $\mathbf{G}$ is commutative, we say that $\mathbf{G}$ is a **commutative group**, or an **abelian group** after the Norwegian mathematician Niels Abel (1802-1829).

Problem 2. *Which of the binary operations* $+,-,\cdot,\cap,\cup,\sim,\circ,+_{12},*,\bullet$ *are associative? For each one that is not, give a counterexample to show it is not.*

Problem 3. *Which of the binary operations* $+,-,\cdot,\cap,\cup,\sim,\circ,+_{12},*,\bullet$ *are commutative? For each one that is not, give a specific counterexample to show that it is not. Reviewing Problem 1, how can you tell from looking at a table if an operation is commutative?*

Problem 4. *What is the identity of each of the groups* $\mathbf{K}_4$*,* $\mathbf{Z}_{12}$*,* $\mathbf{P}_{abc}$ *and* $\mathbf{S}_3$*? List the elements of each group and, under each element, list its inverse.*

Problem 5. *Show that the product of two members of* $\mathbf{M}_1$ *is again a member of* $\mathbf{M}_1$*, and that* $\mathbf{M}_1$ *forms a group under matrix multiplication.*

Problem 6. *Let* **L** *be the binary system on the right. Is* **L** *commutative? Is it a group?*

$*$	0	1	2	3	4
0	0	1	2	3	4
1	1	0	3	4	2
2	2	3	4	1	0
3	3	4	0	2	1
4	4	2	1	0	3